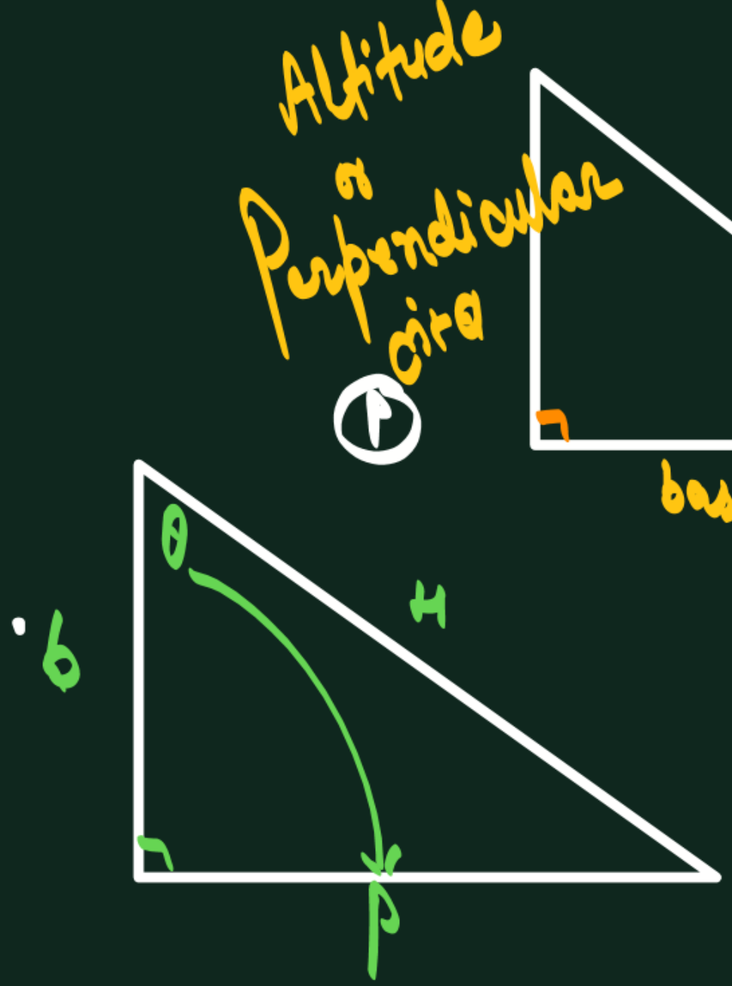
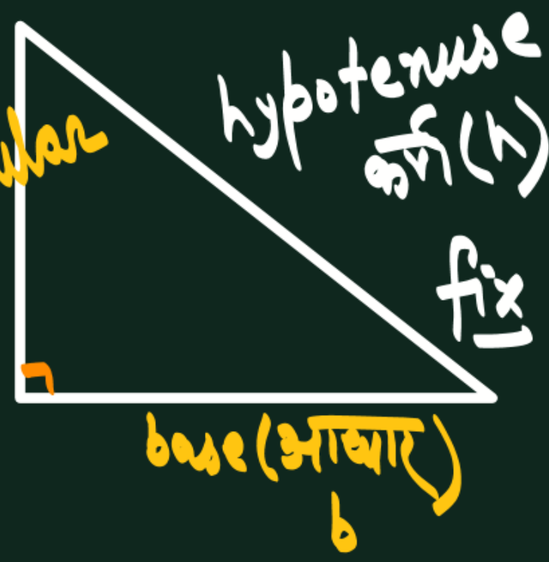
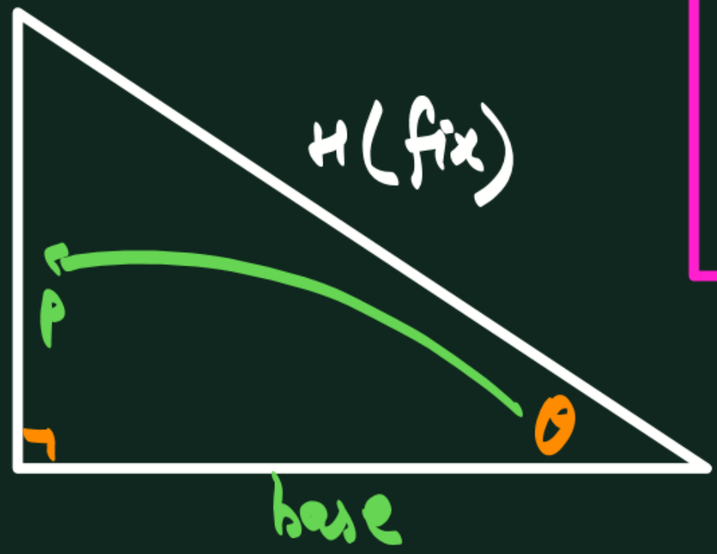
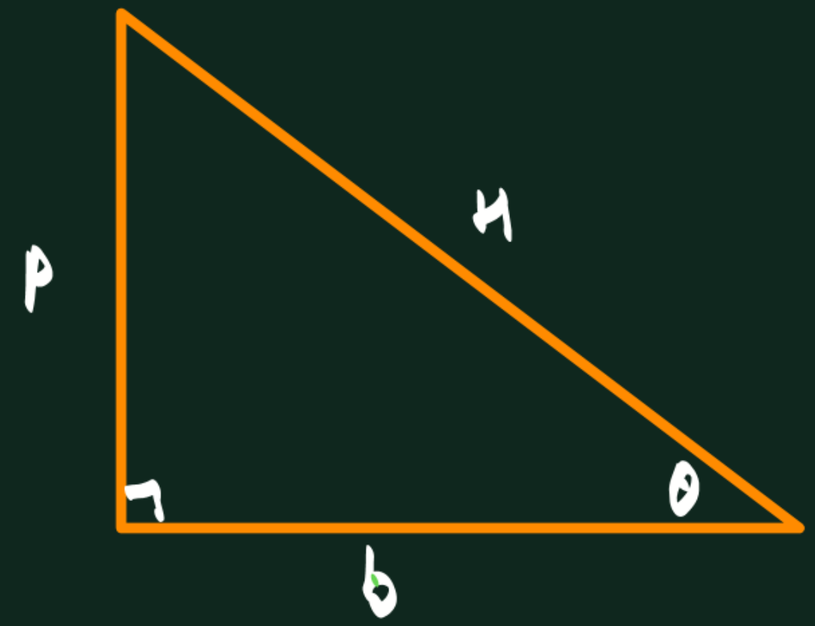
Height and distance
ऊँचाई और दूरी







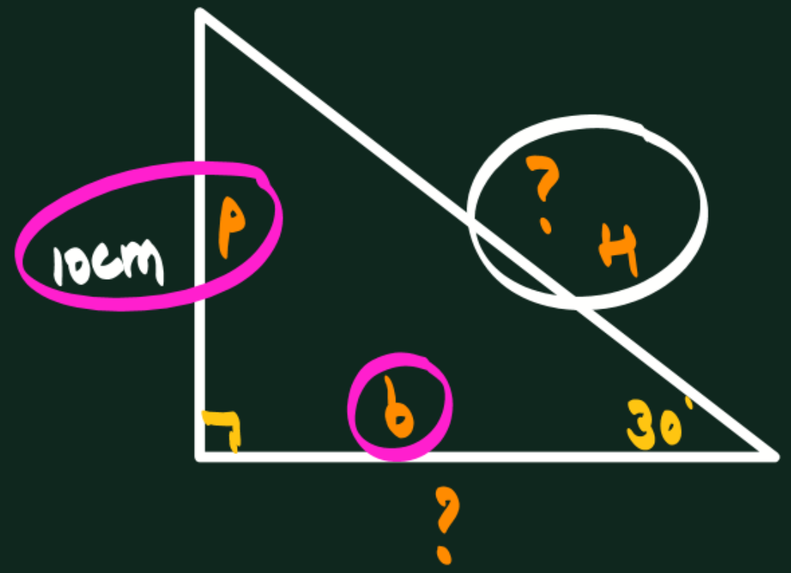
1. θ (4-5)
2. θ (1-2)
3. θ (1-2)



$\sin\theta = \frac{P}{H} = \frac{\text{लंब}}{\text{कर्ण}}$

$\cos\theta = \frac{b}{H} = \frac{\text{आधार}}{\text{कर्ण}}$

$\tan\theta = \frac{P}{b} = \frac{\text{लम्ब}}{\text{आधार}}$ ✓



$\tan 30 = \frac{10}{b}$

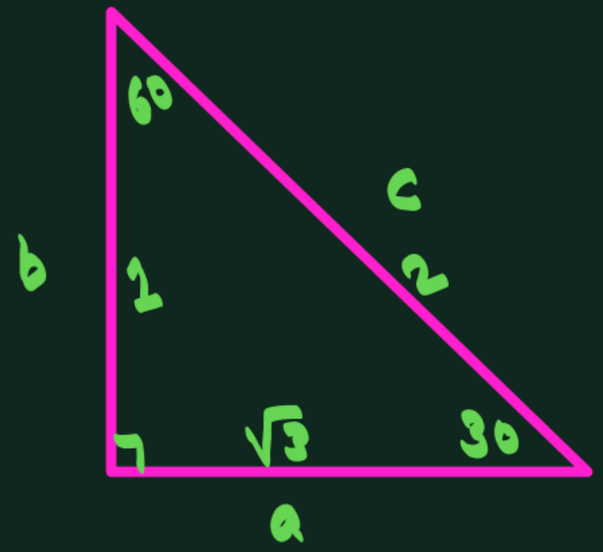
$\frac{1}{\sqrt{3}} = \frac{10}{b}$

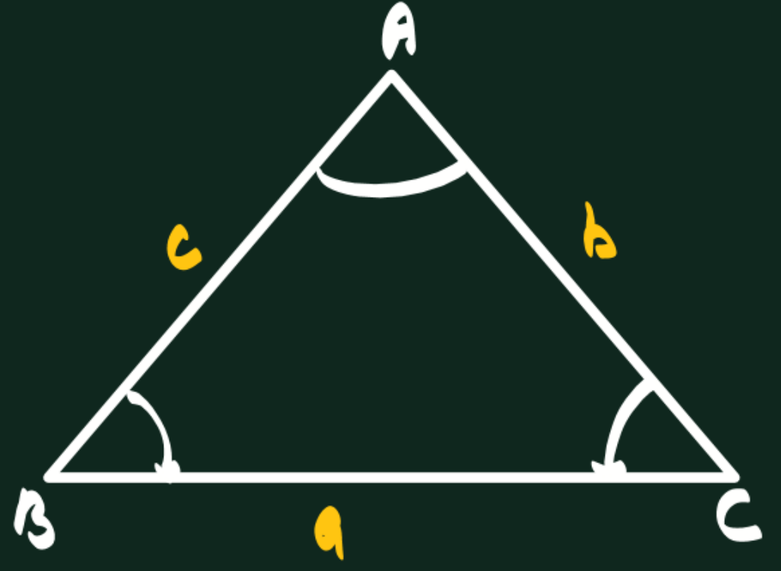
$b = 10\sqrt{3}$

$\sin 30 = \frac{10}{H}$

$\frac{1}{2} = \frac{10}{H}$

$H = 20$



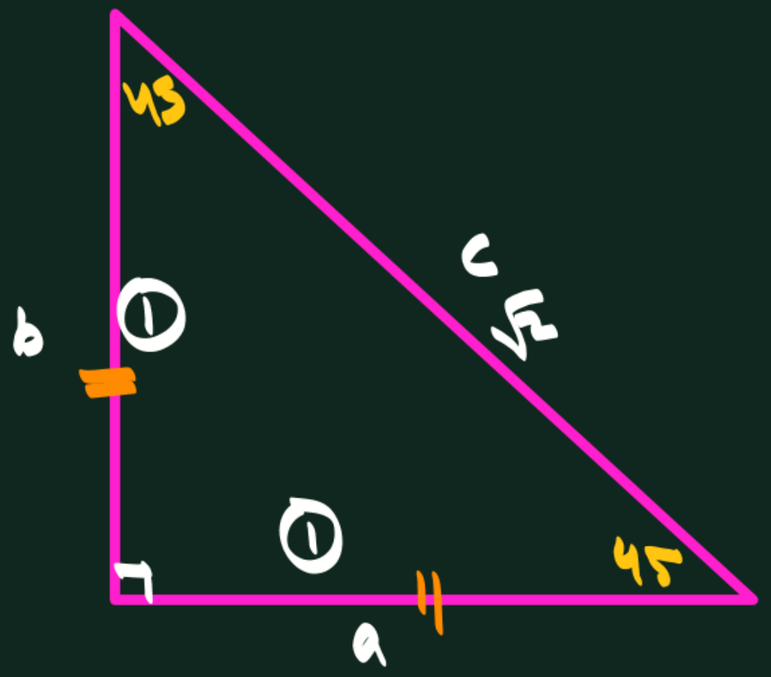


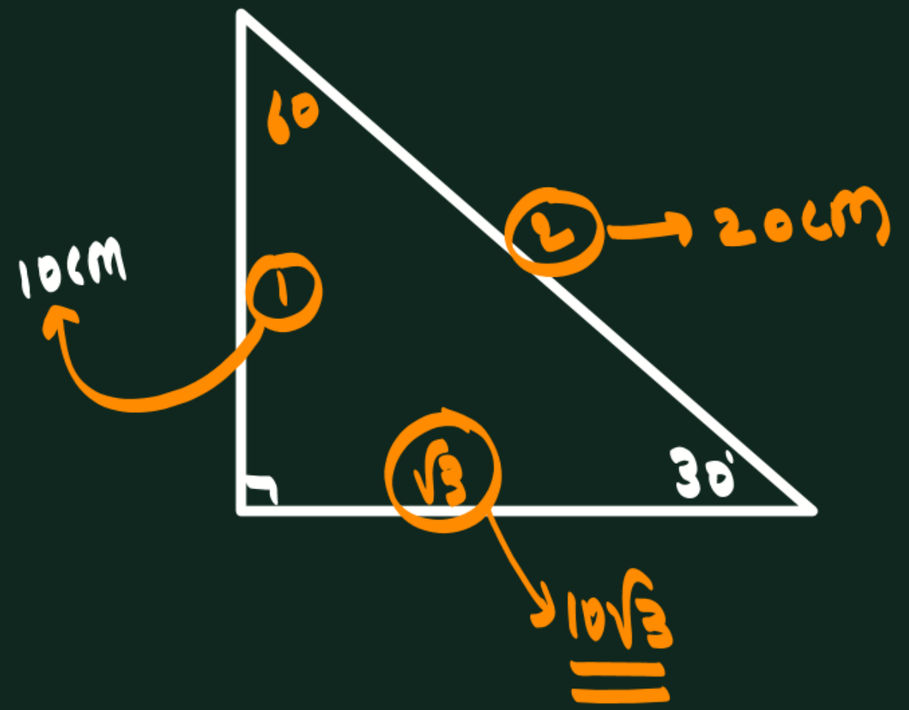
$$a:b:c = \sin 60 : \sin 30 : \sin 90$$

$$= \left(\frac{\sqrt{3}}{2} : \frac{1}{2} : \frac{1}{1}\right) \times 2$$

$$= \sqrt{3} : 1 : 2$$

$$a:b:c = \sin A : \sin B : \sin C$$



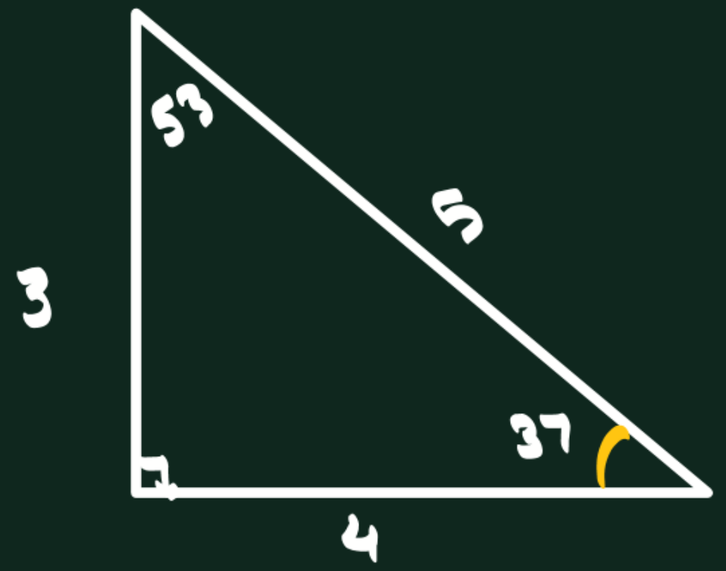


$$a : b : c = \text{Sinus} : \text{Sinus} : \text{Sin}\,90$$

$$= \left(\frac{1}{\sqrt{2}} : \frac{1}{\sqrt{2}} : \frac{1}{1}\right) \times \sqrt{2}$$

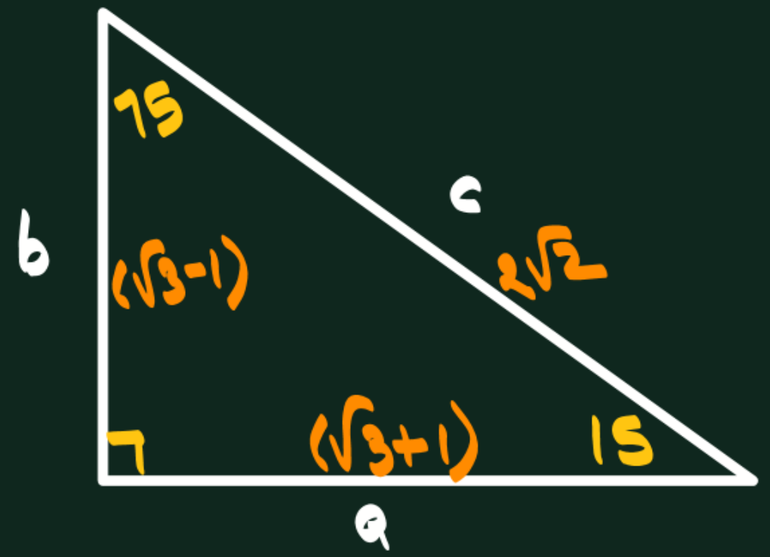
$$= 1 : 1 : \sqrt{2}$$

(*)



$\tan 37 = \frac{3}{4} \rightarrow \frac{P}{b}$

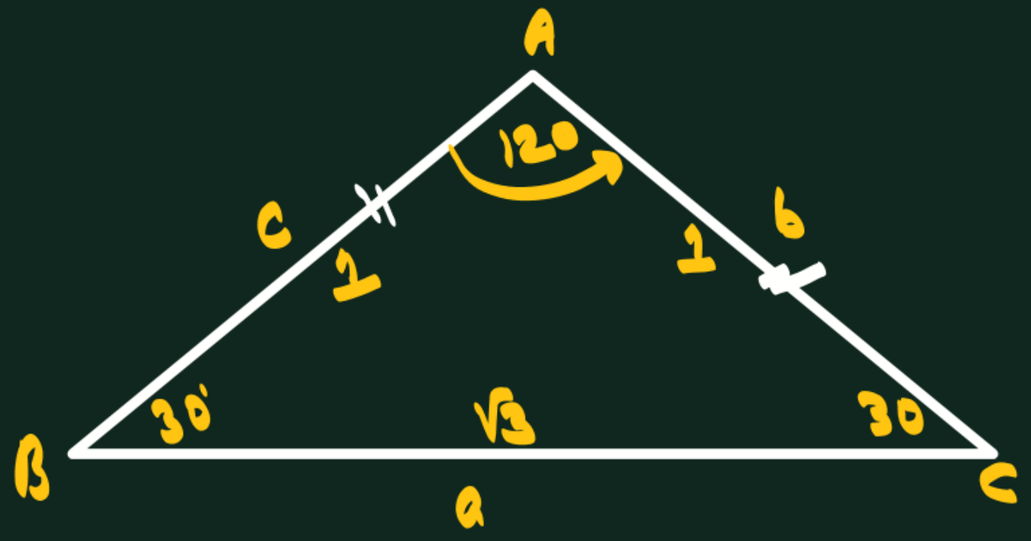
$H = 5$



$a : b : c \rightarrow \sin 75 : \sin 15 : \sin 90$

$\rightarrow \left(\frac{\sqrt{3}+1}{2\sqrt{2}} : \frac{\sqrt{3}-1}{2\sqrt{2}} : 1 \right) \times 2\sqrt{2}$

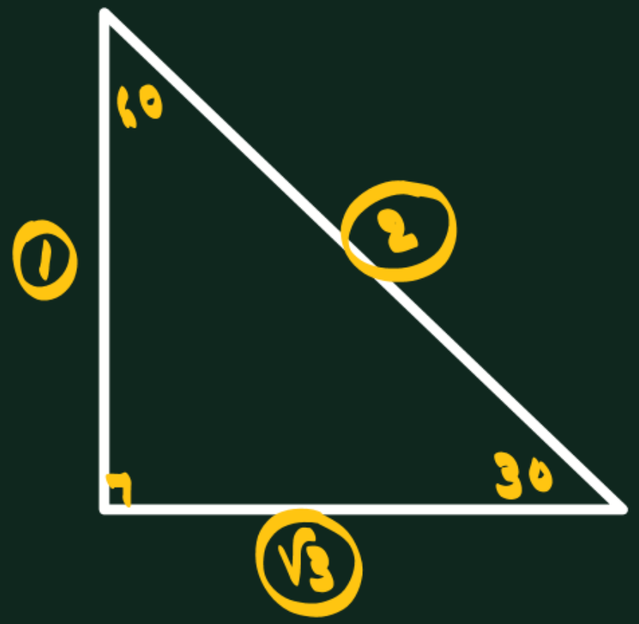
$\Rightarrow (\sqrt{3}+1) : (\sqrt{3}-1) : 2\sqrt{2}$

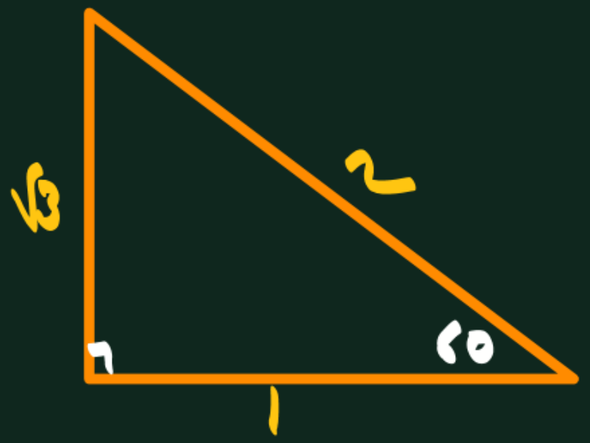


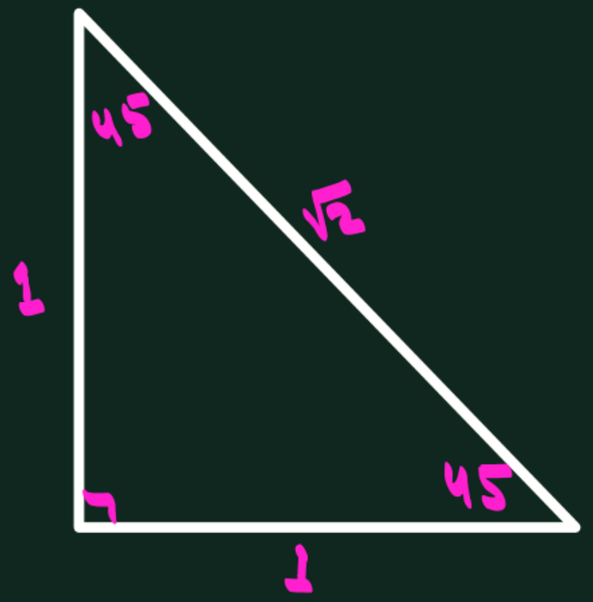
$$a : b : c = \sin 120 : \sin 30 : \sin 30$$

$$= \frac{\sqrt{3}}{2} : \frac{1}{2} : \frac{1}{2}$$

$$\Rightarrow \sqrt{3} : 1 : 1$$

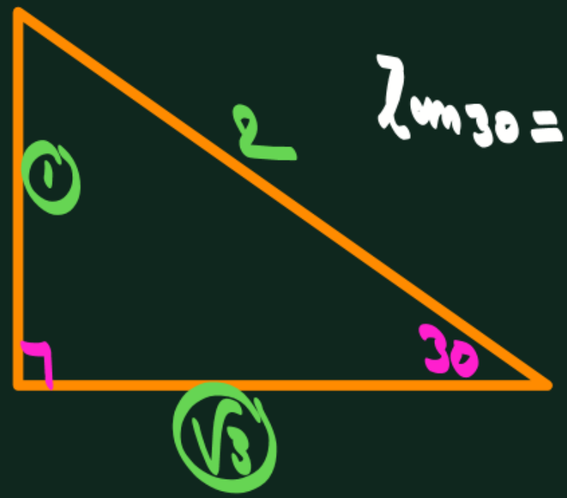




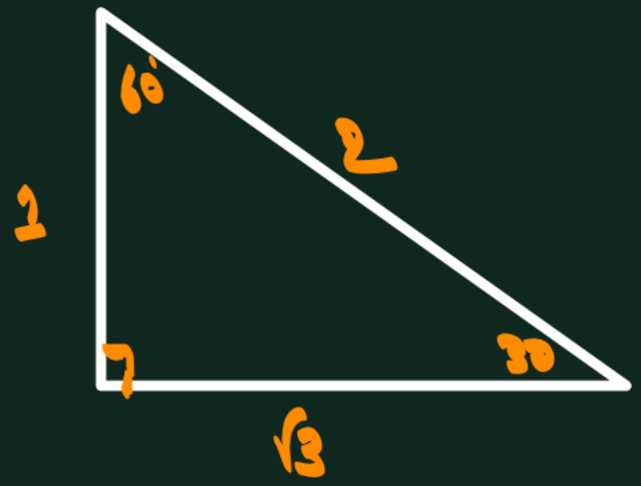


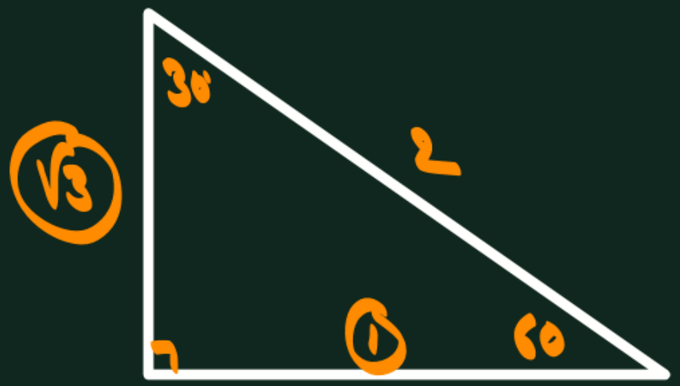
$$\tan 60 = \frac{\sqrt{3}}{1} = \frac{P}{b}$$

$$h = 2$$

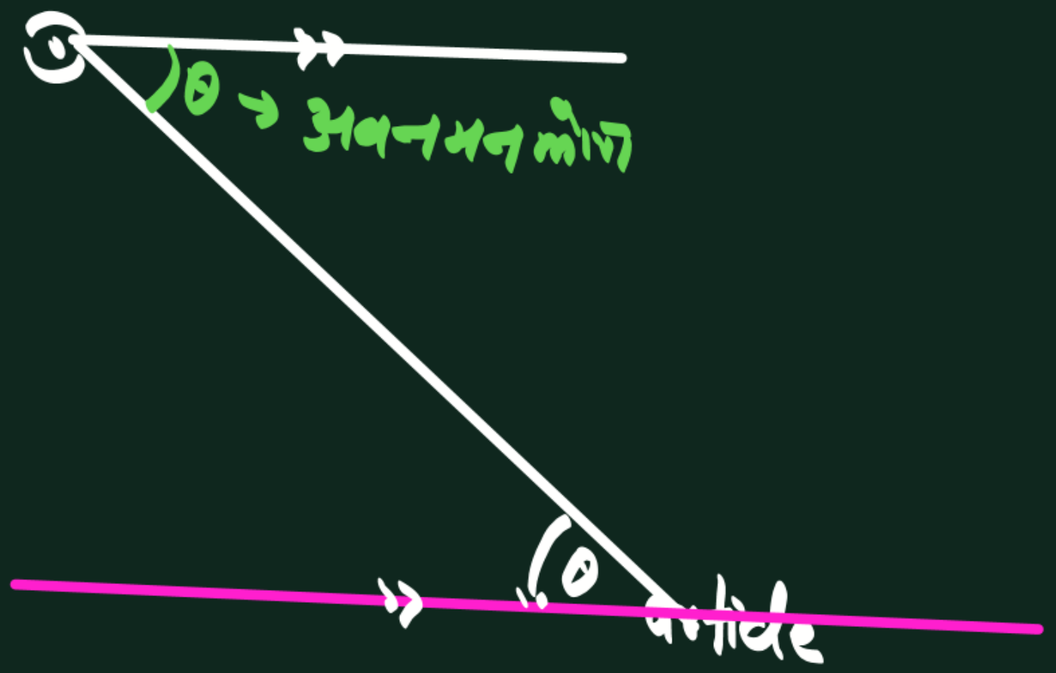


$$\tan 30 = \frac{1}{\sqrt{3}} = \frac{P}{b}$$

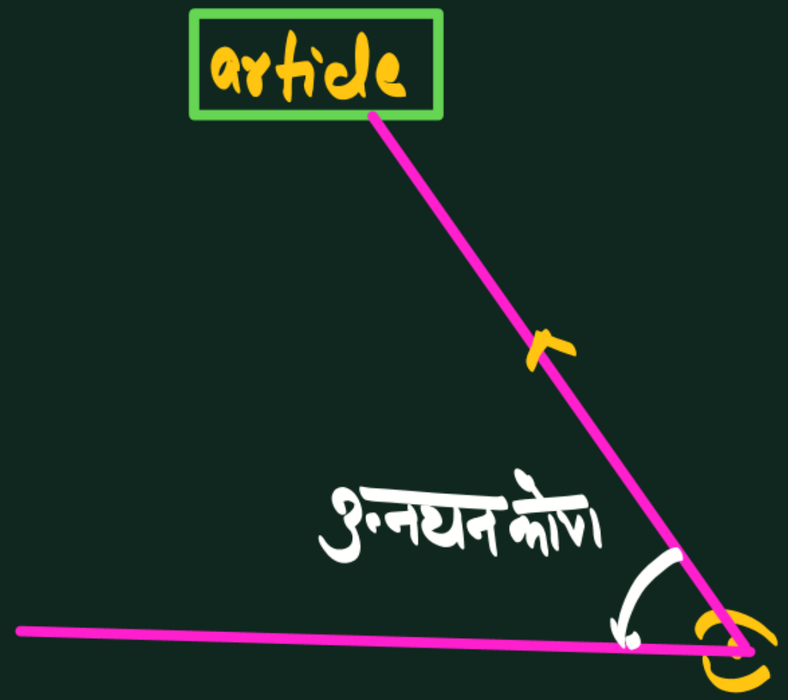

60°
2
1
90°
√3


30°
2
√3
1
60°

① Angle of depression
अवनमन कोण / अवनति कोण



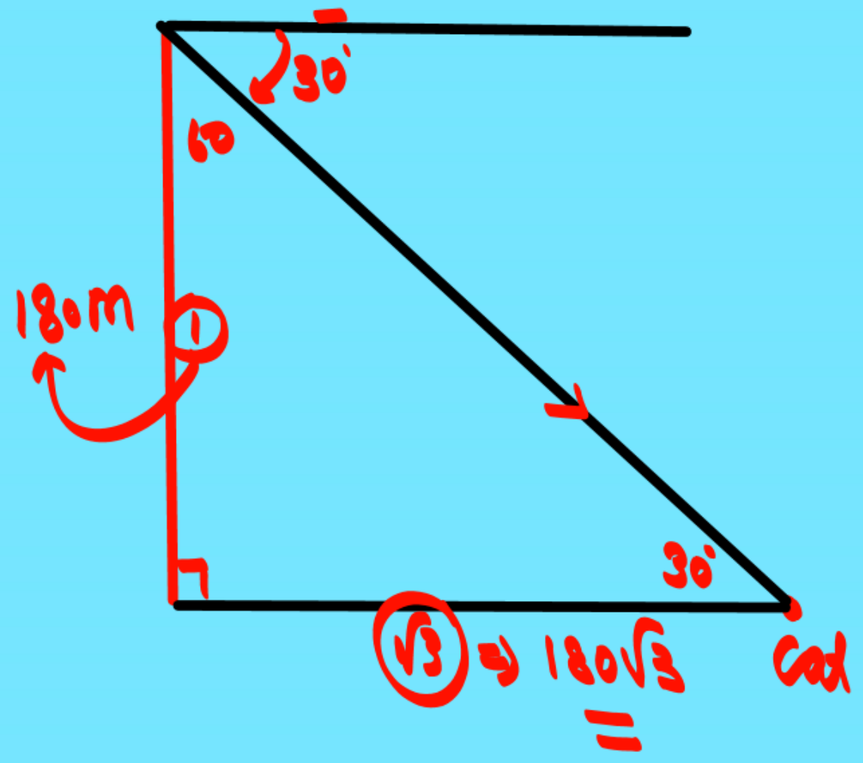
② उन्नयन कोण / उन्नति कोण
उत्क्रांत / Angle of Elevation




Top/शिखर
Bottom/जड़ foot/अधोभाग

HEIGHT & DISTANCE

TYPE–I

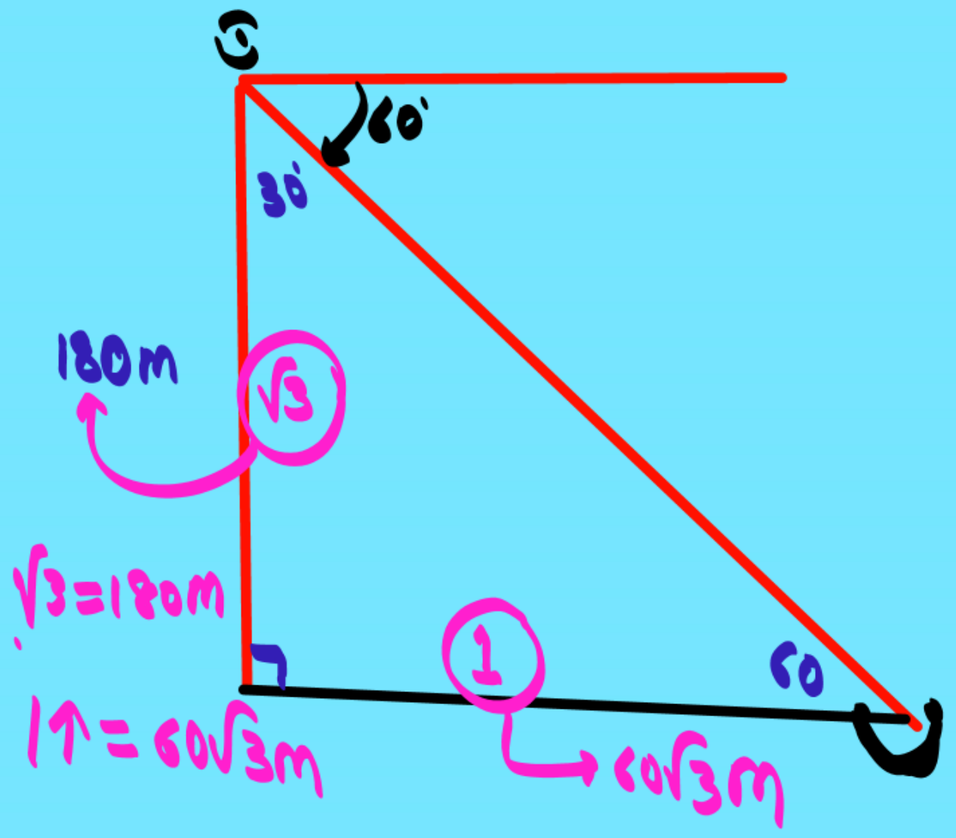


1. From the top of a tower 180 m high, it was observed that the angle of depression of a cat sitting on the ground was 30°. Find the distance of the cat from the foot of the tower.

एक 180 मीटर ऊँचे टावर के शिखर से देखने पर एक बिल्ली जो जमीन पर बैठी है, का अवनमन कोण 30° पाया गया। बिल्ली टावर के नीचले सिरे से कितने दूरी पर बैठी थी ?

(a) 180 m
(b) $180\sqrt{3}$ m
(c) 170 m
(d) $170\sqrt{3}$ m

HEIGHT & DISTANCE

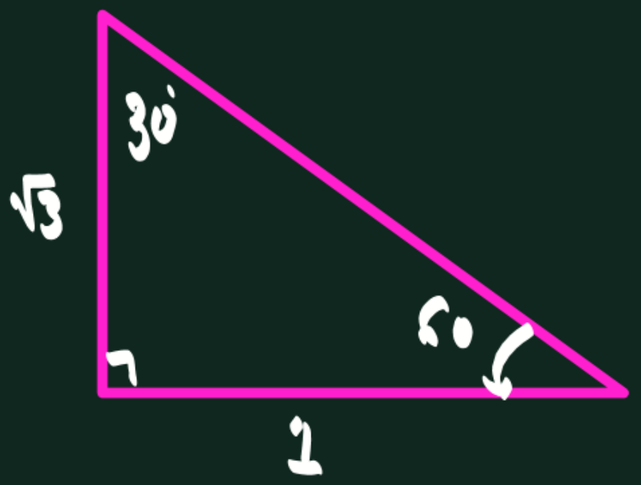


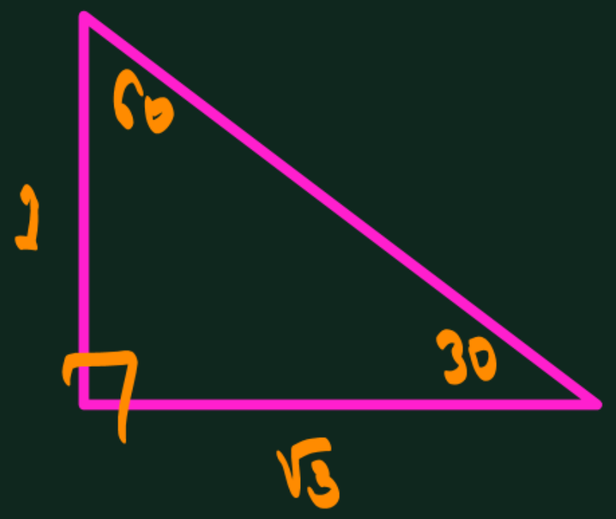
180m / √3 → 60×3 / √3 → 60×√3×√3 / √3 → 60√3

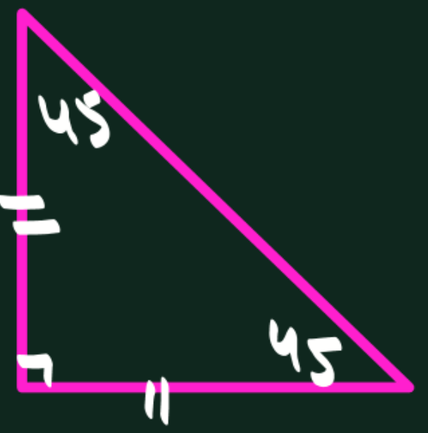
2. A guard observes an enemy boat, from an observation tower at a height of 180 metre above sea level to be at an angle of depression of 60°. The distance of the boat from the foot of the observation tower is ?

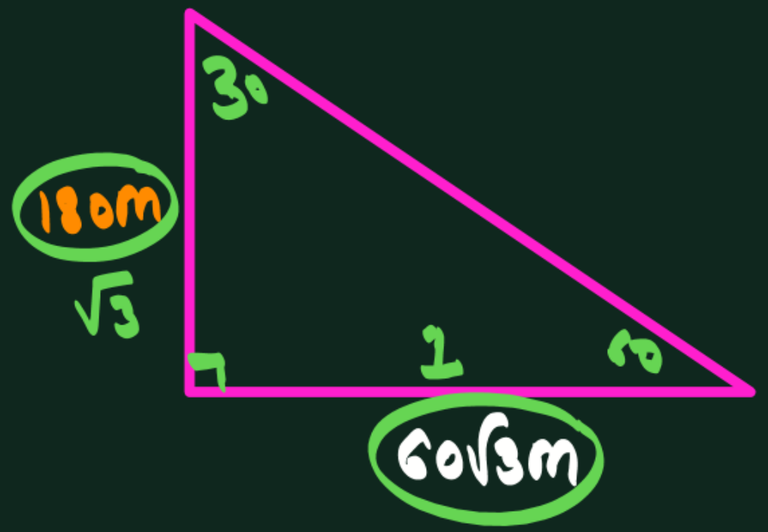
यदि समुद्री-तल से 180 मीटर की ऊँचाई पर स्थित एक अन्वेषण टावर के ऊँपरी भाग से एक गॉर्ड एक शत्रु बोट (नाव) को 60° अवनमन कोण पर देखता है, तो अन्वेषण टॉवर के तल से बोट (नाव) की दूरी क्या होगी ?

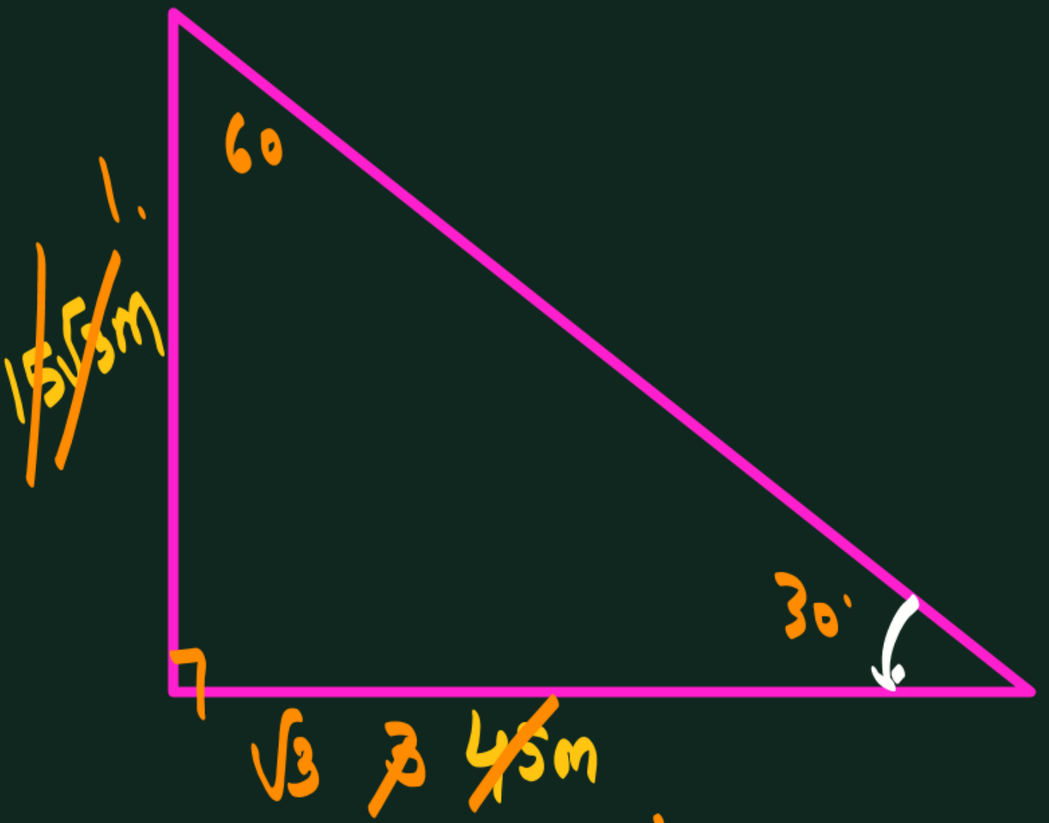
(a) 180 m
(b) $180\sqrt{3}$ m
(c) $60\sqrt{3}$ m
(d) 60 m


30
√3
60
1


60
1
30
√3


45
1
45
1





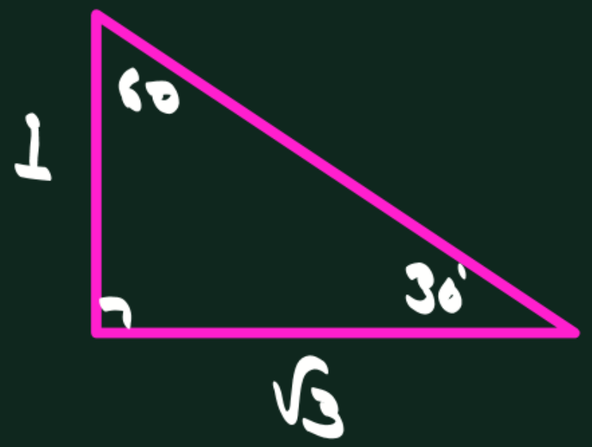
$180 : 60\sqrt{3}$

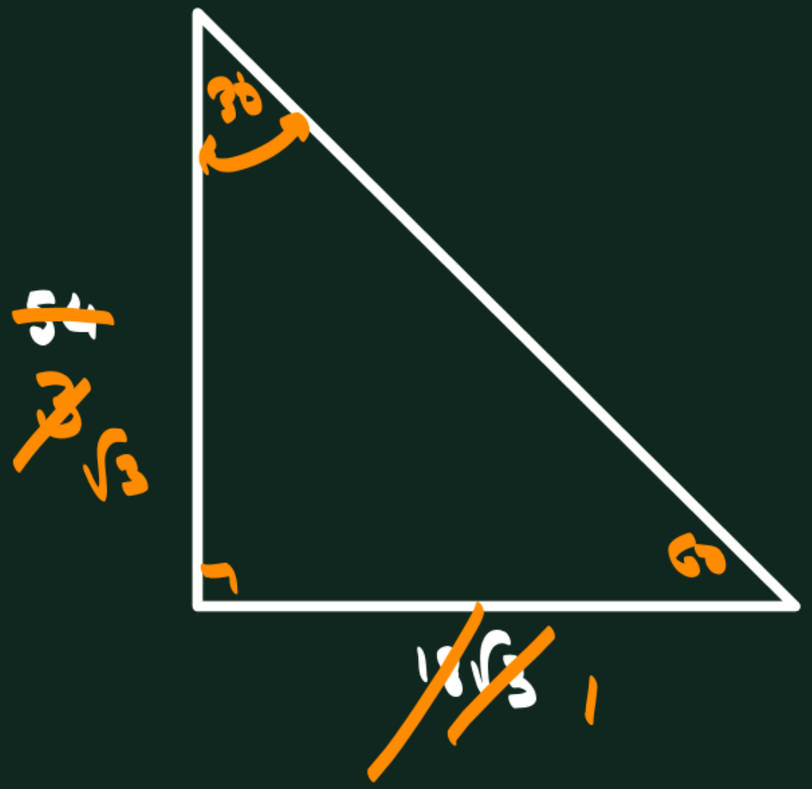
$3 : \sqrt{3}$

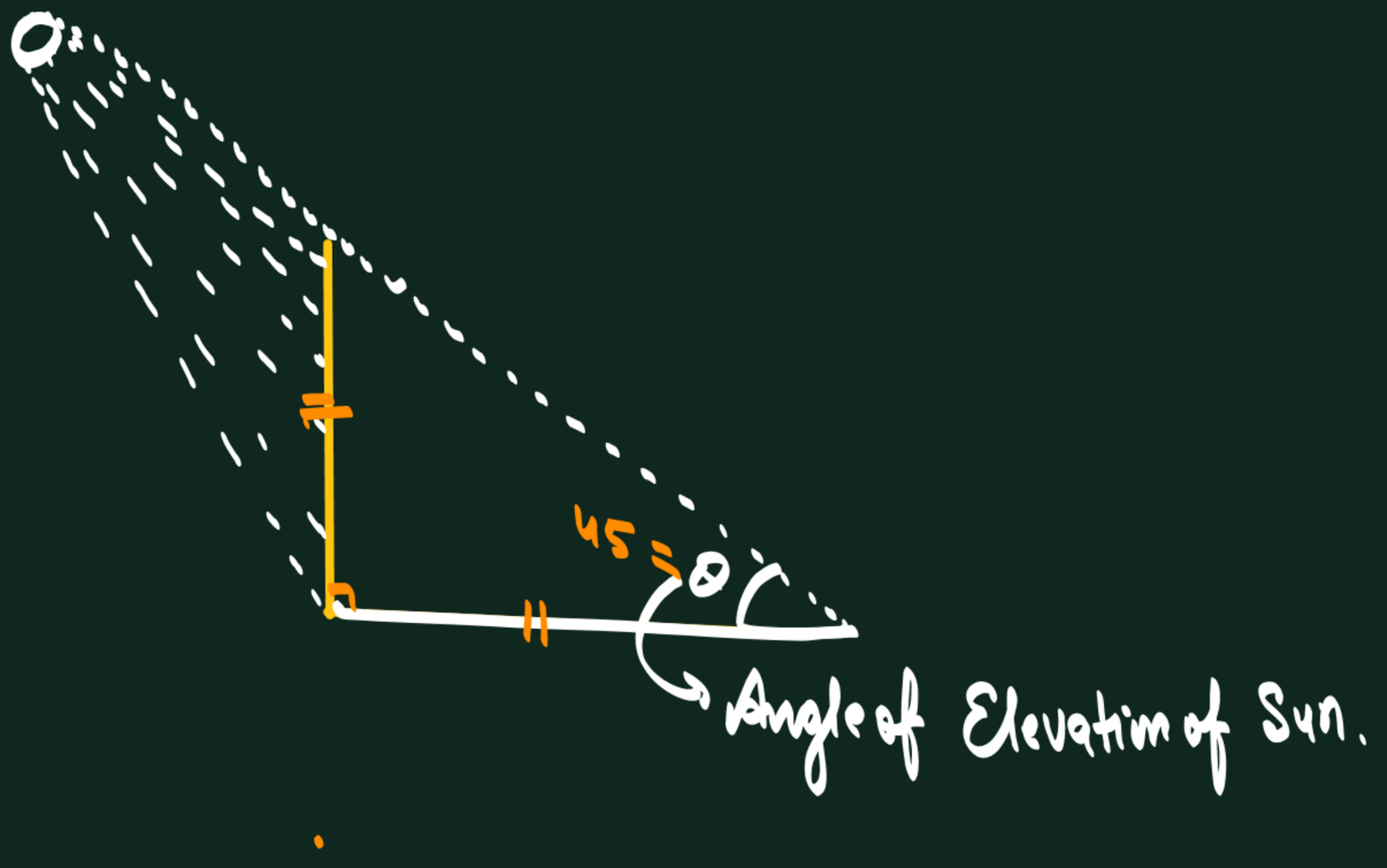
$\sqrt{3} : 1$

ধাম্বা = $h \times \sqrt{3}$

$\frac{\text{ধাম্বা}}{h} = \frac{\sqrt{3}}{1}$






45°
Angle of Elevation of Sun.